

Problem 1 (H3170234) not really sure where divisibility comes in

We claim the solutions are,

- $f(x) = -x + c$, which obviously works.
- $f(x) = \begin{cases} -x + c_1 & \text{for even } x \\ -x + c_2 & \text{for odd } x \end{cases}$ for even numbers c_1, c_2 , which works too case-working y 's parity.

Claim: f is an involution

Plugging in $x = y = 0$,

$$f(f(0)) + f(f(0)) = 0 \Leftrightarrow f(f(0)) = 0$$

Plugging in $x = 0$, we are done.

Therefore,

$$\begin{aligned} f(2x + f(y)) + f(f(2x)) &= y \\ \Leftrightarrow f(2x + f(y)) &= y - 2x \\ \Leftrightarrow f(f(2x + f(y))) &= f(y - 2x) \\ \Leftrightarrow 2x + f(y) &= f(y - 2x) \\ \Leftrightarrow 2x &= f(y - 2x) - f(y) \quad (*) \end{aligned}$$

Plug in $x = -a; y = 0$ and let $f(0) = c$,

$$(*) \Rightarrow -2a + c = f(2a)$$

Plug in $x = -a; y = 1$, let $f(1) = d$.

$$(*) \Rightarrow -2a + d = f(2a + 1)$$

Therefore,

$$f(x) = \begin{cases} -x + c_1 & \text{for even } x \\ -x + c_2 & \text{for odd } x \end{cases}$$

If c_1, c_2 have distinct parity, then $f(f(x))$ must have a different parity than x .

Case 1: c_1 and c_2 are both odd.

Plugging in an odd x ,

$$f(f(x)) = x \Leftrightarrow x - c_2 + c_1 = x$$

Therefore, $c_1 = c_2$. This is our first solution.

Case 2: c_1 and c_2 are both even.

This is our second solution.

Problem 2 (21SLN1)

We claim the only solution is $n = 2$, accomplished by, for example, $a = b = 2$.

We know,

$$a^2 + b + 3 \mid ab + 3b + 8$$

Subtracting the right-hand side from $(a + 3)(a^2 + b + 3)$,

$$\Leftrightarrow a^2 + b + 3 \mid a^3 + 3a^2 + 3a + 1$$

$$\Leftrightarrow a^2 + b + 3 \mid (a + 1)^3$$

However, since $a^2 + b + 3$ isn't the divisor of the cube of any prime,

$$\Leftrightarrow a^2 + b + 3 \mid (a + 1)^2$$

$$\Leftrightarrow a^2 + b + 3 \mid 2a + 1 - b - 3$$

For $a > 2$, we have,

$$|2a + 1 - b - 3| \leq |2a + 1| + |b + 3| < a^2 + b + 3$$

For $a = 2$, we have,

$$|2a + 1 - b - 3| = |2 - b| < b + 7 = a^2 + b + 3$$

For $a = 1$, we have,

$$|2a + 1 - b - 3| = b < b + 4 = a^2 + b + 3$$

Therefore, all cases tell us that,

$$2a + 1 - b - 3 = 0 \Leftrightarrow b = 2a - 2$$

We plug this into the equation for n ,

$$n = \frac{a(2a - 2) + 3(2a - 2) + 8}{a^2 + (2a - 2) + 3} = \frac{2a^2 + 4a + 2}{a^2 + 2a + 1} = 2$$

5 clubs; 5.5 hours (incl. examples)

Problem 3 (98IMO4)

We claim that the set of solutions are,

1. All solutions of the form $(x, y) = (7k^2, 7k)$ for positive integers k . We'd get $\frac{x^2y+x+y}{xy^2+y+7} = k$ so this set of solutions works.
2. $(x, y) = (11, 1)$ and $(x, y) = (49, 1)$.

We are given,

$$xy^2 + y + 7 \mid x^2y + x + y$$

$$\begin{aligned} \Rightarrow xy^2 + y + 7 &| x^2y^2 + xy + y^2 \\ \Leftrightarrow xy^2 + y + 7 &| y^2 - 7x \end{aligned}$$

Case 1: $y^2 - 7x = 0 \Leftrightarrow x = \frac{y^2}{7}$.

This would require $y \mid 7$ for x to be integer, so we write $y = 7k$,

$$\Leftrightarrow x = 7k^2$$

This gives us the set of solutions that goes $(x, y) = (7k^2, 7k)$.

Case 2: $|y^2 - 7x| \geq |xy^2 + y + 7|$

Subcase 1: $y^2 \geq 7x$

$$\Leftrightarrow y^2 - 7x \geq xy^2 + y + 7 \geq y^2 + 7$$

This is impossible.

Subcase 2: $y^2 < 7x$

$$\Leftrightarrow 7x - y^2 \geq xy^2 + y + 7$$

$$\Leftrightarrow 7x \geq (x + 1)y^2 + y + 7$$

Note that if $y^2 \geq 7$; $7x < (x + 1)y^2$ and things become impossible.

If $y = 1$,

$$x + 8 \mid x^2 + x + 1$$

$$\Leftrightarrow x + 8 \mid x^2 + x + 1 - (x + 8)(x - 7)$$

$$\Leftrightarrow x + 8 \mid 57$$

Therefore, the only solutions are $x = 11, 49$.

If $y = 2$,

$$4x + 9 \mid 2x^2 + x + 2$$

$$\Rightarrow 4x + 9 \mid 4x^2 + 2x + 4$$

$$\Leftrightarrow 4x + 9 \mid -7x + 4$$

$$\Leftrightarrow 4x + 9 \mid -28x + 16$$

$$\Leftrightarrow 4x + 9 \mid 79$$

Clearly, nothing works for positive x .

We have found all our solutions.

Problem 4 (17HMMTA4)

We claim the only solutions are $(a, b) = (2, 2^{2017})$; $(a, b) = (1, 1)$, which both obviously work.

We let $d := \gcd(a, b)$. We let $A = \frac{a}{d}$; $B = \frac{b}{d}$.

$$\begin{aligned}
 & ab \mid a^{2017} + b \\
 \Leftrightarrow & d^2 AB \mid d^{2017} A^{2017} + dB \\
 \Leftrightarrow & dAB \mid d^{2016} A^{2017} + B \\
 \Rightarrow & A \mid d^{2016} A^{2017} + B \\
 \Leftrightarrow & A \mid B
 \end{aligned}$$

However, A and B are relatively prime, so $A = 1$. We then get $d = a$; $b = Bd = Ba$. Hence,

$$ab \mid a^{2017} + b$$

Clearly, $a \mid b$, we let $b_1 = \frac{b}{a}$,

$$\Leftrightarrow ab_1 \mid a^{2016} + b_1$$

Clearly, $a \mid b_1$, we now let $b_2 = \frac{b_1}{a} = \frac{b}{a^2}$,

$$\Leftrightarrow ab_2 \mid a^{2015} + b_2$$

We may do this recursively until, for $b_{2017} = \frac{b}{a^{2017}}$,

$$\Leftrightarrow ab_{2017} \mid 1 + b_{2017}$$

Obviously, $0 < 1 + b_{2017} \leq ab_{2017} + ab_{2017}$.

Case 1: $ab_{2017} = 1 + b_{2017} \Leftrightarrow (a - 1)b_{2017} = 1$, giving us $a = 2$; $b_{2017} = 1 \Leftrightarrow b = 2^{2017}$.

Case 2: $2ab_{2017} = 1 + b_{2017} \Leftrightarrow (2a - 1)b_{2017} = 1$, giving us $a = 1$; $b_{2017} = 1 \Leftrightarrow b = 1$.

These are the solutions we expected.

9 clubs; 7 hours

Problem 6 (05SLN6)

Note that,

$$a^n + n \mid b^n + n$$

This means $a < b$, in addition,

$$\Leftrightarrow a^n + n \mid b^n - a^n$$

We fix a and prove that $a = b$. For any prime p , we can use CRT to get that we may select a number n such that $n \equiv 1 \pmod{p-1}$; $n \equiv -a \pmod{p}$. For this selection of n , $p \mid a^n + n$. Therefore,

$$\Rightarrow p \mid b - a$$

FTSOC, let $b > a$. We find a prime p greater than $b - a$ and get a contradiction. Therefore, $a = b$.

Problem 8 (13ESLN4)

We claim $a = b = 1; c = 2$, which obviously works, is the only solution.

Note that,

$$\begin{aligned} n + c &\mid a^n + b^n + n \\ \Leftrightarrow n + c &\mid a^n + b^n - c \quad (*) \end{aligned}$$

Claim: $c = a + b$

Note that for any prime $p > 2013$, we have an integer n which satisfies,

$$\begin{aligned} n &\equiv -c \pmod{p}; \\ n &\equiv 1 \pmod{p-1}; \\ n &\equiv 1 \pmod{p_i} \text{ for all primes } p_i \leq 2013 \end{aligned}$$

This is because all primes are relatively prime to each other, and any primes $p_i \leq 2013$ that $p - 1$ can be a multiple of we already have $n \equiv 1 \pmod{p_i}$. Since n is obviously relatively prime to all primes less than 2013, we can plug n to get,

$$(*) \Rightarrow p \mid a + b - c$$

For a sufficiently large p , we get the RHS must be zero so $c = a + b$.

Claim: $a = b = 1$

From the previous claim, we have,

$$(*) \Leftrightarrow n + a + b \mid a^n + b^n - a - b$$

Note that by Dirichlet's theorem on arithmetic progressions there are infinitely many primes that are $2 \pmod{3}$. For any such $p > 2013$, there exists an integer n which satisfies,

$$\begin{aligned} n &\equiv -a - b \pmod{p}; \\ n &\equiv 3 \pmod{p-1}; \\ n &\equiv 3 \pmod{p_i} \text{ for all primes } p_i \leq 2013 \text{ and } p_i \neq 3; \\ n &\equiv 1 \text{ or } 2 \pmod{3} \end{aligned}$$

The first three conditions can be accomplished the same as above, then the fourth condition can be added using Chinese remainder theorem. (What's a better way of proving this exists?)

We plug in this selection of n to have,

$$\Rightarrow p \mid a^3 + b^3 - a - b$$

For a sufficiently large p , we would get that,

$$a^3 + b^3 - a - b = 0$$

$$\Leftrightarrow a^3 + b^3 = a + b$$

By bounding, this requires $a = b = 1$. We then get $c = a + b = 2$.

17 clubs; 8 hours

Problem 10 (11SLN3)

We claim the answer is all $f(x) = \pm x^d + c$ for some integer constant c and d a divisor of n .

$$f(x) - f(y) = \pm(x^d - y^d) \mid x^n - y^n$$

Obviously works for $d \mid n$.

Plugging in $x = 0$,

$$f(x) - f(0) \mid x^n$$

We let $g(x) = f(x) - f(0)$. We get,

$$g(x) \mid x^n$$

Plugging any prime p into x ,

$$g(p) \mid p^n$$

$$\Leftrightarrow g(p) \in (\pm 1, \pm p, \pm p^2, \dots, \pm p^n)$$

Plugging in $x = 1; y = 0$, we get that $x \in \{\pm 1\}$ as well.

WLOG, we let $g(1) = 1$.

We plug in $x = p$ is a prime number, $y = 1$,

$$\pm p^a + 1 \mid p^n - 1$$

By polynomial division with $F(p) := p^n - 1$ and treating $\pm p^a + 1$ as a root, combined with the fact that n is odd, we must have $g(p) = p^a$ for $a \mid p$. (Do I need to explain this further?)

We now fix y and plug in $x = p$ into the original condition. For any primes p ,

$$p^a - g(y) \mid p^n - y^n$$

$$\Leftrightarrow p^a - g(y) \mid (g(y))^{\frac{n}{a}} - y^n$$

Since $g(y) \mid y^n \Rightarrow |g(y)| \leq y^n$, we can plug in a sufficiently large prime p , we get, for a fixed $a \mid n$,

$$(g(y))^{\frac{n}{a}} - y^n = 0 \Rightarrow g(y) = y^a \forall y \in \mathbb{Z}$$

Having $g(1) = -1$ produces the analogous result that $g(y) = -y^a$.

Since $g(x) = \pm x^a$ for $a \mid n$, we have $f(x) = \pm x^d + c$ for some $d \mid n$ and integer c .

Problem 12 (16SLN6)

We claim that $f(x) = x^2$, which obviously works as $(m^2 - mn + n^2)(m + n) = m^3 + n^3$.

We plug in, $\forall p$ where p is prime or 1, $m = n = p$,

$$\begin{aligned} 2f(p) - p^2 &\mid 2pf(p) \\ \Leftrightarrow 2f(p) - p^2 &\mid p^3 \end{aligned}$$

Since $f(p)$ is positive, we get that,

$$\begin{aligned} 2f(p) - p^2 &\in \{-p, -1, 1, p, p^2, p^3\} \\ \Leftrightarrow f(p) &\in \left\{ \frac{p^2 - p}{2}, \frac{p^2 + 1}{2}, \frac{p^2 + p}{2}, p^2, \frac{p^3 + p^2}{2} \right\} \end{aligned}$$

When $p = 1$, we get that $f(1) = 1$.

Plugging in $m = p, n = 1$ for some prime p ,

$$\begin{aligned} f(p) + 1 - p &\mid pf(p) + 1 \\ \Leftrightarrow f(p) + 1 - p &\mid p(p - 1) + 1 = p^2 - p + 1 \\ \Leftrightarrow 2f(p) + 2 - 2p &\mid 2p^2 - 2p + 2 \end{aligned}$$

Case 1: $f(p) = \frac{p^2 - p}{2}$ meaning $p^2 - 3p + 2 \mid 2p^2 - 2p + 2 \Leftrightarrow p^2 - 3p + 2 \mid 4p - 2$. If $p = 2$, then the LHS would be 0. If $p = 3, 5$, this obviously doesn't work. By bounding, this does not work for $p \geq 7$.

Case 2: $f(p) = \frac{p^2 - 1}{2}$ meaning $p^2 - 2p + 1 \mid 2p^2 - 2p + 2 \Leftrightarrow p^2 - 2p + 1 \mid 2p$. If $p = 2$, then $f(p)$ does not return an integer. $p = 3$ doesn't work. We prove that $p \geq 3$ cannot work by bounding.

Case 3: $f(p) = \frac{p^2 + 1}{2}$ meaning $p^2 - 2p + 3 \mid 2p^2 - 2p + 2 \Leftrightarrow p^2 - 2p + 3 \mid 2p - 4$. If $p = 2$, then $f(p)$ does not return an integer. We prove that $p \geq 3$ cannot work by bounding.

Case 4: $f(p) = \frac{p^2 + p}{2}$ meaning $p^2 - p + 2 \mid 2p^2 - 2p + 2 \Leftrightarrow p^2 - p + 2 \mid -2$ by bounding impossible.

Case 5: $f(p) = \frac{p^3 + p^2}{2}$ meaning $p^3 + p^2 - 2p + 2 \mid 2p^2 - 2p + 2$. By bounding impossible.

Therefore, **we have** $f(p) = p^2$ since all others are impossible.

We now fix n . For any prime p , we plug in $m = p$ to get,

$$\begin{aligned}
& p^2 + f(n) - np \mid p(p^2) + nf(n) \\
& \Leftrightarrow p^2 + f(n) - np \mid p(np - f(n)) + nf(n) \\
& \Leftrightarrow p^2 + f(n) - np \mid n(np - f(n)) - pf(n) + nf(n) \\
& \Leftrightarrow p^2 + f(n) - np \mid -pf(n) + n^2p
\end{aligned}$$

Given a large enough p , the LHS would be greater than the right. Therefore, for all n ,

$$-pf(n) + n^2p = 0 \Leftrightarrow f(n) = n^2$$

27 clubs; 9.5 hours

Problem 14 (11IMO5)

Plugging in $n = 0$, we get,

$$f(m) \mid f(m) - f(0) \Leftrightarrow f(m) \mid f(0) \quad \forall \text{ integer } m$$

Since $f(0)$ is finite and $f(m) \leq f(0)$, we see that f is bounded with $f(0)$ as a maximum. Plugging in $m = 0$, we get that,

$$f(-n) \mid f(0) + f(n) \Leftrightarrow f(-n) \mid f(n) \quad \forall n \in \mathbb{Z}$$

Plugging in $x = -n$, we then see that $f(x) \mid f(-x)$. Therefore, $f(x) = f(-x)$.

We plug in $m = -a; n = b$ to get,

$$f(-a - b) \mid f(a) - f(b)$$

Similarly,

$$f(-b - c) \mid f(b) - f(c);$$

$$f(-c - a) \mid f(c) - f(a)$$

Therefore, for any pairs of integers a, b, c where $a + b + c = 0$,

$$f(c) \mid f(a) - f(b);$$

$$f(a) \mid f(b) - f(c);$$

$$f(b) \mid f(c) - f(a);$$

WLOG, let $f(c)$ be the greatest of $f(a), f(b), f(c)$. Since $f(c) > f(a) - f(b)$, we have,

Conclusion: For any $a + b + c = 0$ where $f(c) \geq f(a), f(b)$, $f(a) = f(b) \mid f(c)$.

Therefore, for any $f(c) \geq f(a)$, we plug in $b = -a - c$. Since,

$$f(-a - c) \mid f(c) - f(a) \Rightarrow f(-a - c) \leq f(c)$$

We use the conclusion to get $f(a) \mid f(c)$ as desired.

Problem 15 (17ELMO1)

We claim that for any $p \geq 3$, v_p of the LHS is less than or equal to v_p of the RHS.

$$v_p(\gcd(a_1^n + P, a_2^n + P, \dots, a_n^n + P)) \leq v_p(2 \gcd(a_1, a_2, \dots, a_n)^n)$$

$$\Leftrightarrow \min(v_p(a_1^n + a_1 a_2 \dots a_n), \dots, v_p(a_n^n + a_1 a_2 \dots a_n)) \leq n \min(v_p(a_1), \dots, v_p(a_n))$$

WLOG let $v_p(a_1) \leq v_p(a_2) \leq \dots \leq v_p(a_n)$.

$$\Leftrightarrow \min(v_p(a_1^n + a_1 a_2 \dots a_n), \dots, v_p(a_n^n + a_1 a_2 \dots a_n)) \leq n v_p(a_1)$$

Case 1: $v_p(a_1) < v_p(a_n)$, then $v_p(a_1^n + a_1 a_2 \dots a_n) = n v_p(a_1)$ so this is true.

Case 2: $v_p(a_1) = v_p(a_2) = \dots = v_p(a_n)$

We let $x := v_p(a_1) = \dots = v_p(a_n)$. We let,

$$b_1 = \frac{a_1}{p^x}; b_2 = \frac{a_2}{p^x}; \dots; b_n = \frac{a_n}{p^x}$$

We show that $\min(v_p(b_1^n + b_1 \dots b_n), \dots, v_p(b_n^n + b_1 \dots b_n)) = 0$. FTSOC, let this not be the case, which would imply that,

$$p \mid b_1^n + b_1 \dots b_n \Leftrightarrow b_1 b_2 \dots b_n \equiv -b_1^n \pmod{p}$$

$$p \mid b_2^n + b_1 \dots b_n \Leftrightarrow b_1 b_2 \dots b_n \equiv -b_2^n \pmod{p}$$

...

$$p \mid b_n^n + b_1 \dots b_n \Leftrightarrow b_1 b_2 \dots b_n \equiv -b_n^n \pmod{p}$$

Multiplying the congruences together, we get,

$$b_1^n \dots b_n^n \equiv -b_1^n \dots b_n^n \pmod{p}$$

Therefore, since p is odd, $b_1^n \dots b_n^n \equiv 0 \pmod{p}$ which contradicts $p \nmid b_1, \dots, b_n$.

Now for $p = 2$, we have must instead show,

$$\min(v_2(a_1^n + a_1 a_2 \dots a_n), \dots, v_2(a_n^n + a_1 a_2 \dots a_n)) \leq n \min(v_2(a_1), \dots, v_2(a_n)) + 1$$

WLOG let $v_2(a_1) \leq v_2(a_2) \leq \dots \leq v_2(a_n)$. Analogously, if $v_2(a_1) < v_2(a_n)$, we are done.

If $x := v_2(a_1) = \dots = v_2(a_n)$, we let,

$$b_1 = \frac{a_1}{2^x}; b_2 = \frac{a_2}{2^x}; \dots; b_n = \frac{a_n}{2^x}$$

We show that $\min(v_p(b_1^n + b_1 \dots b_n), \dots, v_p(b_n^n + b_1 \dots b_n)) = 1$. FTSOC, let this not be the case.

$$4 \mid b_1^n + b_1 \dots b_n;$$

$$\dots$$

$$4 \mid b_n^n + b_1 \dots b_n;$$

Similarly, as above, we'd get,

$$b_1^n b_2^n \dots b_n^n \equiv -b_1^n b_2^n \dots b_n^n \pmod{4}$$

This means that $b_1^n b_2^n \dots b_n^n \equiv 0, 2 \pmod{4}$, which contradicts all being odd.

Since we have proven that $v_p(\gcd(a_1^n + P, a_2^n + P, \dots, a_n^n + P)) \leq v_p(2 \gcd(a_1, a_2, \dots, a_n)^n)$ for all primes p , $\gcd(a_1^n + P, a_2^n + P, \dots, a_n^n + P) \leq 2 \gcd(a_1, a_2, \dots, a_n)^n$.

32 clubs; 11.5 hours

Problem 17 (ZAB34B8B)

Multiplying the three conditions, we have,

$$a \mid bc + b + c \Leftrightarrow a \mid ab + bc + ca + a + b + c$$

The same can be said for b, c . Since a, b, c are prime,

$$\Leftrightarrow abc \mid ab + bc + ca + a + b + c$$

Since the RHS must be positive,

$$\Rightarrow abc \leq ab + bc + ca + a + b + c \quad (*)$$

WLOG, let $a < b < c$. Note that, since $b \geq a + 1; c \geq b + 1$,

$$bc \geq (a + 1)(b + 1) > (a + 1)(b + 1) - 1 = ab + a + b$$

$$bc \geq bc$$

Since $b \geq a + 1$,

$$bc \geq c(a + 1) = ac + c$$

Adding the three up,

$$3bc > ab + bc + ca + a + b + c$$

Therefore, we cannot have $a \geq 3$ or there'd be a contradiction with (*). This means $a = 2$.

$$(*) \Leftrightarrow 2bc \leq 2b + bc + 2c + 2 + b + c$$

$$\Leftrightarrow bc \leq 3b + 3c + 2$$

If $b = 3$, then,

$$6c \mid 6 + 3c + 2c + 2 + 3 + c$$

$$\Leftrightarrow 6c \mid 11$$

This is impossible.

If $b = 5$, then,

$$\begin{aligned} 10c &| 10 + 5c + 2c + 5 + 2 + c \\ &\Leftrightarrow 10c | 17 + 8c \end{aligned}$$

$c = 7$ doesn't work. By bounding this is impossible for $c \geq 11$.

If $b \geq 7$, then obviously.

$$bc \geq 7c > 3b + 3c + 2$$

Problem 18 (20TSTST4)

We claim the solutions are $(a, b) = (1, 1)$ or $(1, 2)$ or $(2, 1)$.

Note that if $d := \gcd(a, b)$, then,

$$\begin{aligned} a &| b^4 + 1 \\ &\Rightarrow d | 1 \end{aligned}$$

Therefore, a, b are relatively prime.

If $a = b$, $a = b = 1$, one of our solutions.

WLOG, let $b > a$. We let positive integer $k := b - a$. We have,

$$\begin{aligned} a &| (a + k)^4 + 1 \Leftrightarrow a | k^4 + 1 \\ a + k &| a^4 + 1 \Leftrightarrow a + k | k^4 + 1 \end{aligned}$$

Therefore, since $\gcd(a, a + k) = 1$,

$$\Leftrightarrow \mathbf{ab} | \mathbf{k^4 + 1}$$

Claim 1: $\frac{k^4+1}{ab} < 16$

$$\begin{aligned} \lfloor \sqrt{a} \rfloor &= \lfloor \sqrt{b} \rfloor \\ &\Rightarrow \sqrt{b} < \sqrt{a} + 1 \\ &\Leftrightarrow b < a + 2\sqrt{a} + 1 \\ &\Leftrightarrow b \leq a + 2\sqrt{a} \\ &\Leftrightarrow b - a \leq 2\sqrt{a} \end{aligned}$$

This means,

$$k^4 + 1 = (b - a)^4 + 1 \leq 16a^2$$

Therefore,

$$\frac{k^4 + 1}{ab} < \frac{k^4 + 1}{a^2} \leq 16$$

Claim 2: $\frac{k^4+1}{ab}$ is odd.

If a, b have the same parity, the numerator is odd.

If a, b have distinct parities, since $k^4 \equiv 0, 1 \pmod{4}$, the numerator cannot be a multiple of 4. The denominator is a multiple of 2 so it must be odd.

Claim 3: $\frac{k^4+1}{ab}$ has no odd prime divisors.

Since $\frac{k^4+1}{ab}$ is less than 16, we only check odd primes less than 16.

$$k^4 \equiv 0, 1 \pmod{3};$$

$$k^4 \equiv 0, 1 \pmod{5};$$

$$k^4 \equiv 0, 1, 2, 4, 4, 2, 1 \pmod{7};$$

$$k^4 \equiv 0, 1, 5, 4, 3, 9, 9, 3, 4, 5, 1 \pmod{11};$$

$$k^4 \equiv 0, 1, 3, 3, 9, 1, 9, 9, 1, 9, 3, 3, 1 \pmod{13}$$

Clearly, $k^4 + 1$ hence cannot be a multiple of any of these. Therefore, we are done.

From *claims 1, 2*, we have,

$$\frac{k^4 + 1}{ab} = 1$$

$$\Leftrightarrow k^4 + 1 = a(a + k)$$

$$\Leftrightarrow a^2 + ka - (k^4 + 1) = 0$$

Note that for $k > 1$, the discriminant,

$$(2k^2)^2 < k^2 + 4(k^4 + 1) < (2k^2 + 1)^2$$

This means a cannot be an integer for $k > 1$. When $k = 1$, the only positive solution is $a = 1$. This gives the solution $(a, b) = (1, 2)$. Analogously, if $a > b$ instead, we have $(a, b) = (2, 1)$.

39 clubs; 13 hours

Problem 19 (16IMO4)

We claim that $b = 6$. This can be attained with $a = 197$, getting the set,

$$\{197 \cdot 198 + 1, 198 \cdot 199 + 1, 199 \cdot 200 + 1, 200 \cdot 201 + 1, 201 \cdot 202 + 1, 202 \cdot 203 + 1\}$$

We can check that the 1st and 5th term are both multiples of 19, 2nd and 4th both multiples of 7, and 3rd and 6th both multiples of 3.

Note that, for any integers n ,

$$\gcd(P(n), P(n+1)) = \gcd(n^2 + n + 1, n^3 + 3n + 3) = \gcd(n^2 + n + 1, 2n + 2)$$

Since $n^2 + n + 1$ is odd,

$$= \gcd(n(n+1) + 1, n+1) = 1$$

Therefore, $b = 2$ obvious does not work. This also means $b = 3$ doesn't work as $P(a+2)$ would be relatively prime to both $P(a+1)$ and $P(a+3)$.

We may also find that, for any integers n ,

$$\gcd(P(n), P(n+2)) = \gcd(n^2 + n + 1, 4n + 6)$$

Since the LHS is odd,

$$= \gcd(n^2 + n + 1, 2n + 3) = \gcd((n-2)(n+1), 2n+3)$$

We know $\gcd(n+1, 2n+3) = 1$, so,

$$= \gcd(n-2, 2n+3) = \gcd(n-2, 7)$$

Therefore, for $P(n), P(n+2)$ are not relatively prime if and only if $n \equiv 2 \pmod{7}$.

This means that $b = 4$ does not work either. FTSOC, let it work.

$P(a+2)$ is relatively prime with $P(a+1), P(a+3)$ meaning $a \equiv 0 \pmod{7}$ for $P(a+2)$ to not be relatively prime with $P(a+4)$. However, this means $P(a+3)$ is coprime with $P(a+1), P(a+2), P(a+4)$. (!)

Now note that, for any integers n ,

$$\gcd(P(n), P(n+3)) = \gcd(n^2 + n + 1, 6n + 12) = \gcd(n^2 + n + 1, 3n + 6)$$

If $n \equiv 1 \pmod{3}$, then this is at least 3. However, if not, $n^2 + n + 1$ is not divisible by 3.

$$= \gcd(n^2 + n + 1, n+2) = \gcd((n-1)(n+1), n+2) = \gcd(n-1, 3)$$

Since we assumed $n \not\equiv 1 \pmod{3}$, this evaluates to one. Therefore, $P(n), P(n+3)$ are not relatively prime if and only if $n \equiv 1 \pmod{3}$.

We now claim that $b = 5$ does not work. FTSOC, if it does,

From $f(n+3)$ being not relatively prime to either $f(n+1)$ or $f(n+5)$, we get that,

$$n \equiv 1 \text{ or } 6 \pmod{7}$$

Either way, it means that $\gcd(f(n+2), f(n+4)) = 1$. Therefore,

$$\gcd(f(n+2), f(n+5)) > 1;$$

$$\gcd(f(n+1), f(n+4)) > 1$$

This is impossible.

Problem 21 (18RMM4)

(My solution is completely different from the official. It feels shaky but I cannot find any issues.)

If we apply the Euclidean algorithm on with the goal of minimizing the coefficient of n ,

$$\gcd(an + b, cn + d)$$

We'd eventually get, for some integers B, D ,

$$= \gcd(\gcd(a, c)n + B, D)$$

Note that D is non-zero since $ad \neq bc$ as otherwise both $an + b$ and $cn + d$ would be multiples of the same $\gcd(a, c)n + B$ meaning $\frac{a}{b} = \frac{c}{d} \Leftrightarrow ad = bc$ (!).

Note that by the correctness of the Euclidean algorithm, we still have $\gcd(B, D) = \gcd(b, d)$.

Since B and D are both results of adding and subtracting several b, d , so we have, $\gcd(b, d) \mid B, D$. We let, $x := \frac{B}{\gcd(b, d)} = \frac{B}{\gcd(B, D)}$; $Y := \frac{D}{\gcd(b, d)} = \frac{D}{\gcd(B, D)}$. This means $\gcd(x, Y) = 1$.

Plugging in these fixed integers x, Y ,

$$= \gcd(\gcd(a, c)n + x \gcd(b, d), Y \gcd(b, d)) \quad (*)$$

We let $g := \gcd(a, c, Y)$.

Since $\gcd(x, Y) = 1$ and $\gcd(b, d, a, c) = 1$,

$$\gcd(g, \gcd(a, c)n + x \gcd(b, d)) = \gcd(g, x \gcd(b, d)) = 1$$

With this information, we can divide the right term of the $\gcd$ by g .

Therefore, we let $y = \frac{Y}{g}$. This gives us y is relatively prime to $\gcd(a, c)$.

$$(*) = \gcd(\gcd(a, c)n + x \gcd(b, d), y \gcd(b, d))$$

This may only evaluate to positive divisors of $y \gcd(b, d)$. It hence suffices to show that it evaluates to all divisors of $y \gcd(b, d)$ for some values of n . Note that, since $\gcd(\gcd(a, c), y \gcd(b, d)) = 1$,

$$\begin{aligned} \{\gcd(a, c), 2 \gcd(a, c), \dots, y \gcd(b, d)\} &\equiv \{0, 1, 2, \dots, y \gcd(b, d) - 1\} \pmod{y \gcd(b, d)} \\ &\Leftrightarrow \{\gcd(a, c) + x \gcd(b, d), 2 \gcd(a, c) + x \gcd(b, d), \dots, y \gcd(b, d) + x \gcd(b, d)\} \\ &\equiv \{0, 1, 2, \dots, y \gcd(b, d) - 1\} \pmod{y \gcd(b, d)} \end{aligned}$$

Therefore, over $n = 1, 2, \dots, \gcd(a, c)n + x \gcd(b, d)$, we will cover every remainder mod $y \gcd(b, d)$ including all its divisors. We are hence done.

Problem 24 (21HMMTT6)

We claim the answers are, for $n \equiv 1 \pmod{3}$,

$$n = 1 \text{ or } p \text{ or } p^2 \text{ for some prime } p$$

The $n = 1; n = p$ cases are obvious. For $n = p^2$, we see that $d = 1; d = p^2$ are obvious and for $d = p$,

$$\begin{aligned} p^2 + p + 1 &| p^4 + p^2 + 1 \\ \Leftrightarrow p^2 + p + 1 &| (-p - 1)^2 + p^2 + 1 \\ \Leftrightarrow p^2 + p + 1 &| 2p^2 + 2p + 2 \end{aligned}$$

This is true.

Plugging in $d = 1$, we get that $3 \mid n^2 + n + 1 \Leftrightarrow n \equiv 1 \pmod{3}$.

Claim: Any working n has at most 3 divisors.

FTSOC, if n has four divisors, we may write $n = ab$ for $n > a > b > 1$.

$$\begin{aligned} a^2 + a + 1 &| n^2 + n + 1 \\ \Leftrightarrow a^2 + a + 1 &| a^2b^2 + ab + 1 \end{aligned}$$

As we proved earlier, $a^2 + a + 1 \mid a^4 + a^2 + 1$.

$$\Leftrightarrow a^2 + a + 1 \mid a^2b^2 - a^4 + ab - a^2$$

We know that $\gcd(a, a^2 + a + 1) = 1$, so,

$$\begin{aligned} \Leftrightarrow a^2 + a + 1 &| ab^2 - a^3 + b - a \\ \Leftrightarrow a^2 + a + 1 &| (ab + a^2 + 1)(b - a) \\ \Leftrightarrow a^2 + a + 1 &| (ab - a)(b - a) \\ \Leftrightarrow a^2 + a + 1 &| (b - 1)(b - a) \\ \Leftrightarrow a^2 + a + 1 &| b^2 - ab - b + a \end{aligned}$$

We know that,

$$\begin{aligned} a^2 &> ab - b^2; \\ a + 1 &> b - a \end{aligned}$$

Therefore, by bounding,

$$b^2 - ab - b + a = 0 \Leftrightarrow b = 1 \text{ or } b = a$$

Both contradict our hypothesis, so we are done.

50 clubs; 16.5 hours